16 · Case 13, part 1: the kit, ovulation, and the cycle graph

The words of this track, as it says them. Bold brown is the examiner.

Exam case 13. Fertility.

Rose and Simon are seeing the nurse at their GP clinic, because they've decided to start trying to conceive. Rose has been on the Depo Provera injection as contraception for the past two years. As her nurse, you advise that it could take a little while for her normal cycle to settle after coming off this hormonal contraceptive. Rose takes your advice to track her cycle, after learning there are only a few days each month when she's fertile. She has also bought an ovulation predictor kit, and some pregnancy tests.

Her Anatomy Mondays page names her fertility quiz as this case's model answer, so these answers are her quiz keys. And in her session, she said the cycle graph is in the exam, though the labels may be taken away. So know each curve by its shape. LH peaks just before ovulation. Progesterone is low, then peaks in the second half. And FSH goes with the follicular and proliferative phases. Her phrase: keep the F's together.

One mark. What is an ovulation predictor kit? What does it show?

It shows when ovulation is about to happen, because it picks up the sharp peak in luteinising hormone, LH, in the urine, just before ovulation.

Four marks. What occurs during ovulation?

The hook: an LH peak, the follicle bursts, and the oocyte goes out toward the fimbriae.

A secondary oocyte is released from the ovary. Ovulation is triggered by a peak in LH. The ovarian wall ruptures when the fluid-filled Graafian follicle, the tertiary follicle, bursts. And the force of the fluid pushes the oocyte out of the ovary, toward the fimbriae of the fallopian tube.

One mark. Using the image, on what days of her cycle is a female fertile?

Days ten to seventeen.

Two marks. Can you explain why the hormone levels change over the course of the menstrual cycle?

Because each hormone is driving a different process. FSH, for the follicles to develop. Oestrogen, for the uterine lining to thicken. LH, for ovulation. And progesterone, for the uterus to prepare for implantation and pregnancy.

Two marks. How would pregnancy change the hormone levels graph?

The hormone drop at roughly day twenty-one wouldn't happen, with successful implantation and pregnancy. The oestrogen and progesterone levels keep rising.

Once more. Ovulation, and the graph.

The ovulation predictor kit catches the peak in LH in the urine, just before ovulation. At ovulation, an LH peak makes the Graafian follicle burst, the ovarian wall ruptures, and a secondary oocyte is pushed out toward the fimbriae of the fallopian tube. The fertile days: ten to seventeen. Each hormone drives its own process: FSH grows the follicles, oestrogen thickens the lining, LH triggers ovulation, and progesterone prepares for implantation. In pregnancy, the drop at about day twenty-one doesn't happen: oestrogen and progesterone keep rising.